

BMIEG 457 Concept Alternative Presentation

*Project 25: Context-Aware and Efficient Data
Valuation for Medical Machine Learning*

Client

Dr. Rohit Singla | UBC Robotics and Controls Laboratory

Team 17

Jaiden Siu

Chloe Christensen

Dhairya Aggarwal

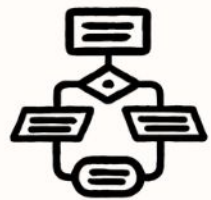
Amy Yu



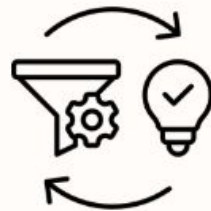
Agenda



1 Project Introduction



2 Function Structure Diagram (FSD)



3 Concept Generation



4 Scoring and Screening Concepts



5 Conclusions and Future Steps

Introduction to Data Valuation

What is Data Valuation

Definition: Data valuation (DV) quantifies how much each training data point contributes to a model's performance.

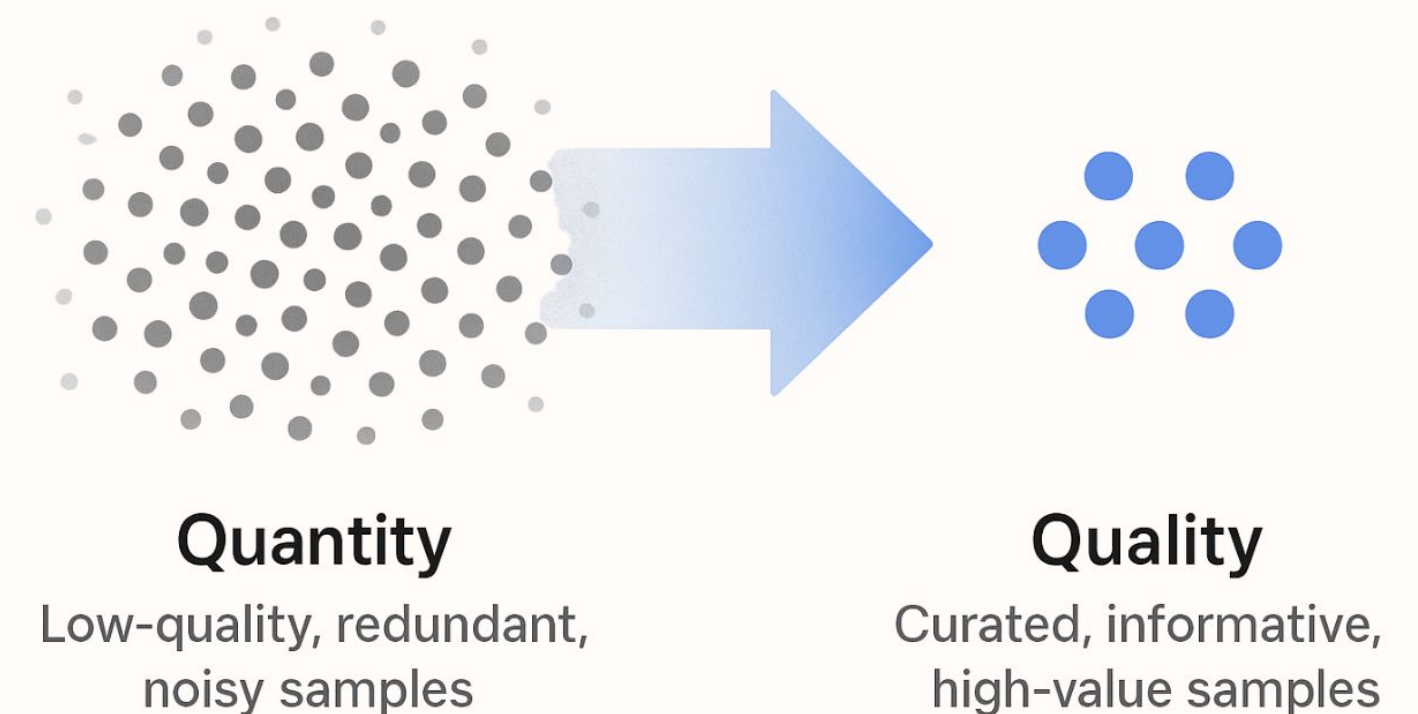
Why It Matters

- Identification of most valuable samples
- Detects redundant or harmful samples that hurt accuracy
- Enables smaller, context-aware datasets → lower compute cost, reduce training time, and decrease carbon emission

How It Works

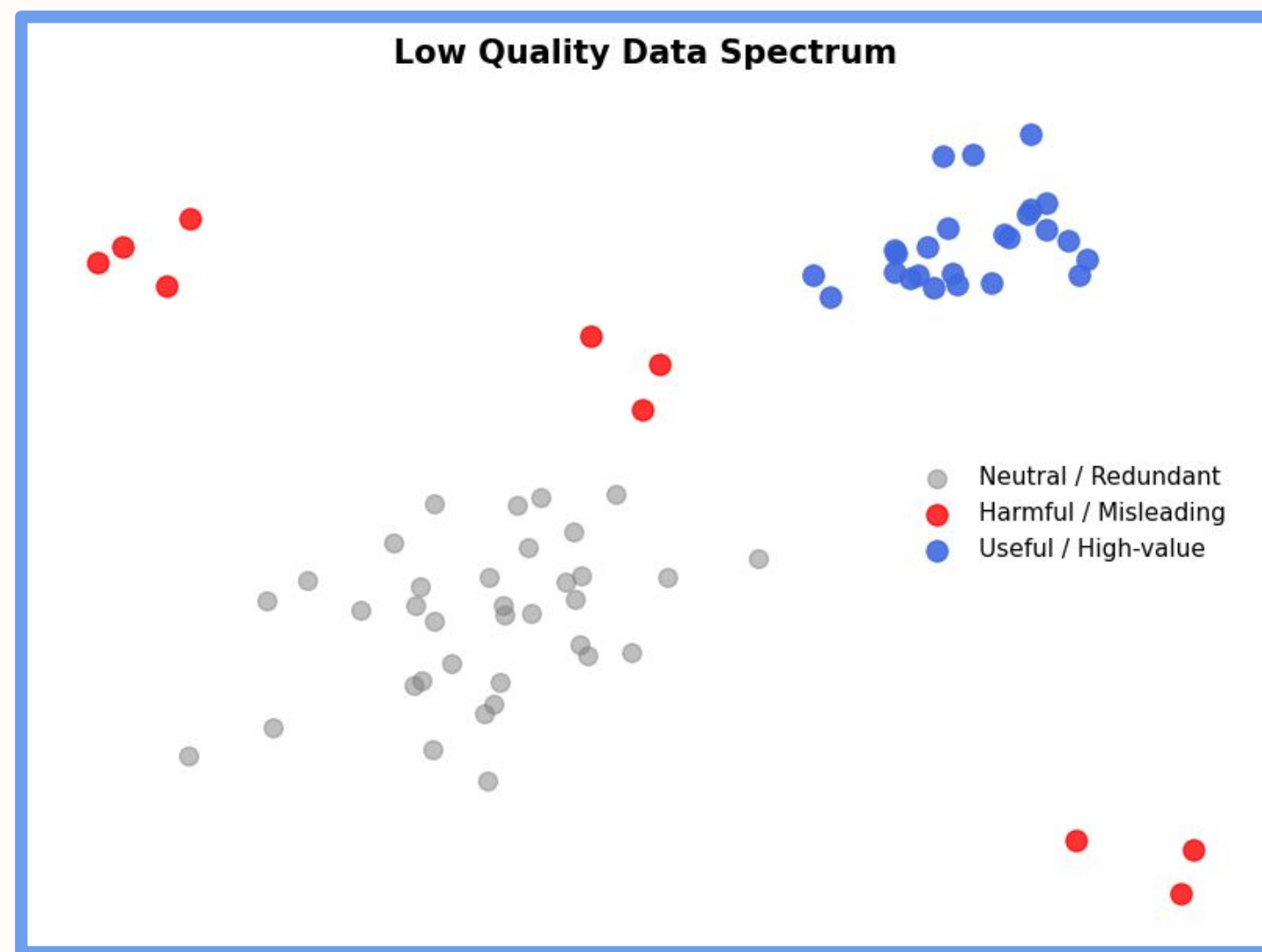
Using methods like Shapley values, fast approximation, etc. to assign each data point a **value score** reflecting its impact on model outcomes during the training phase in a ML pipeline.

From more data is better → better data is smarter



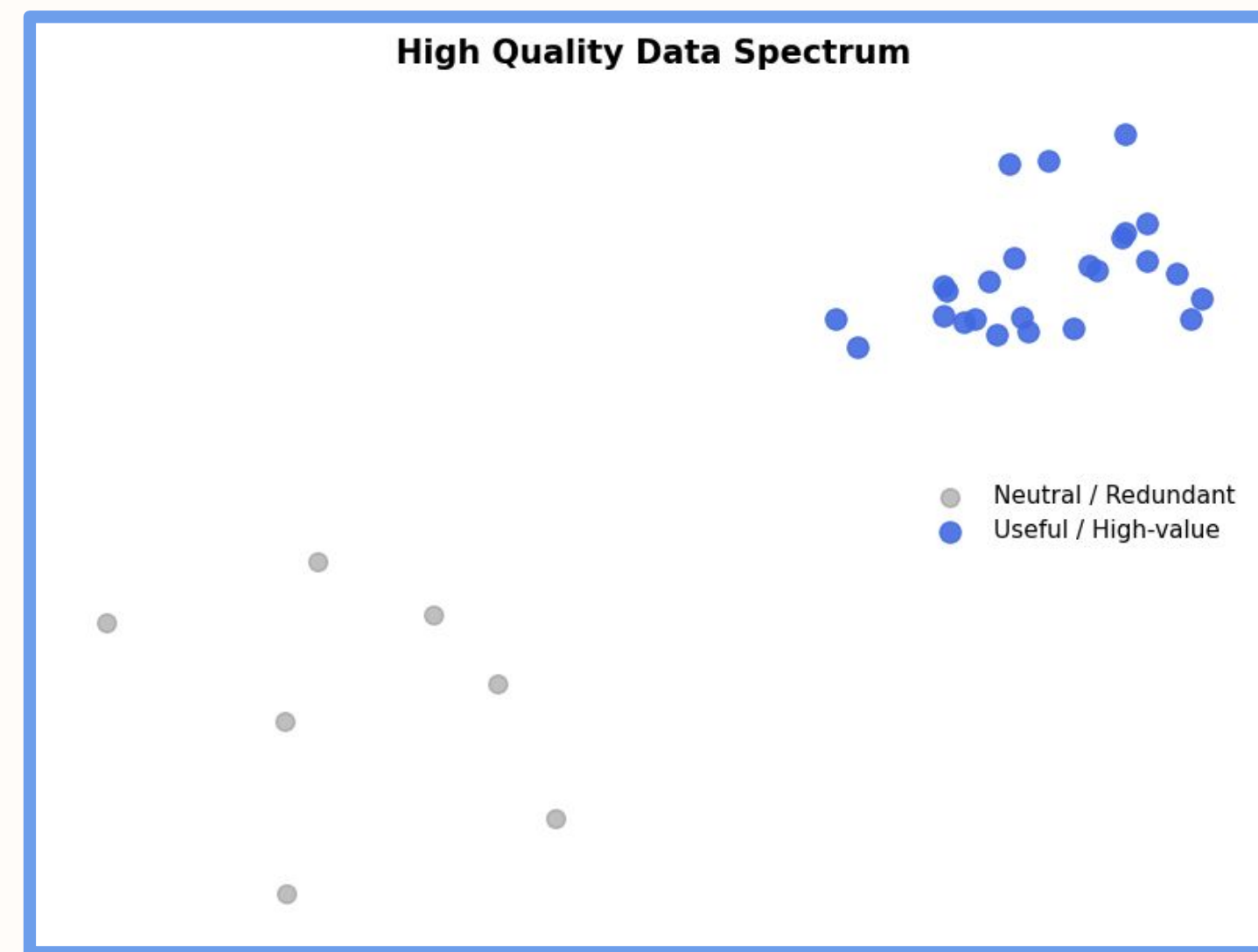
From More Data ➡ To Better Data

Quantity



Low-quality, redundant dataset

Quality



Curated, high quality dataset

Problem

Current Challenges in AI Training

- Expensive (training GPT-4 costed \$100M [1])
- High carbon emissions (training GPT-3 emitted 500 metric tons of CO2 [2])
- Long training time

Current Challenges in DV Research

- Limited benchmarking between DV methods due to different:
 - Integration / evaluation methods
 - Range of modalities
 - Task types
 - ML/DL architectures

[1] [Source 1](#)

[2] [Source 2](#)

Objective

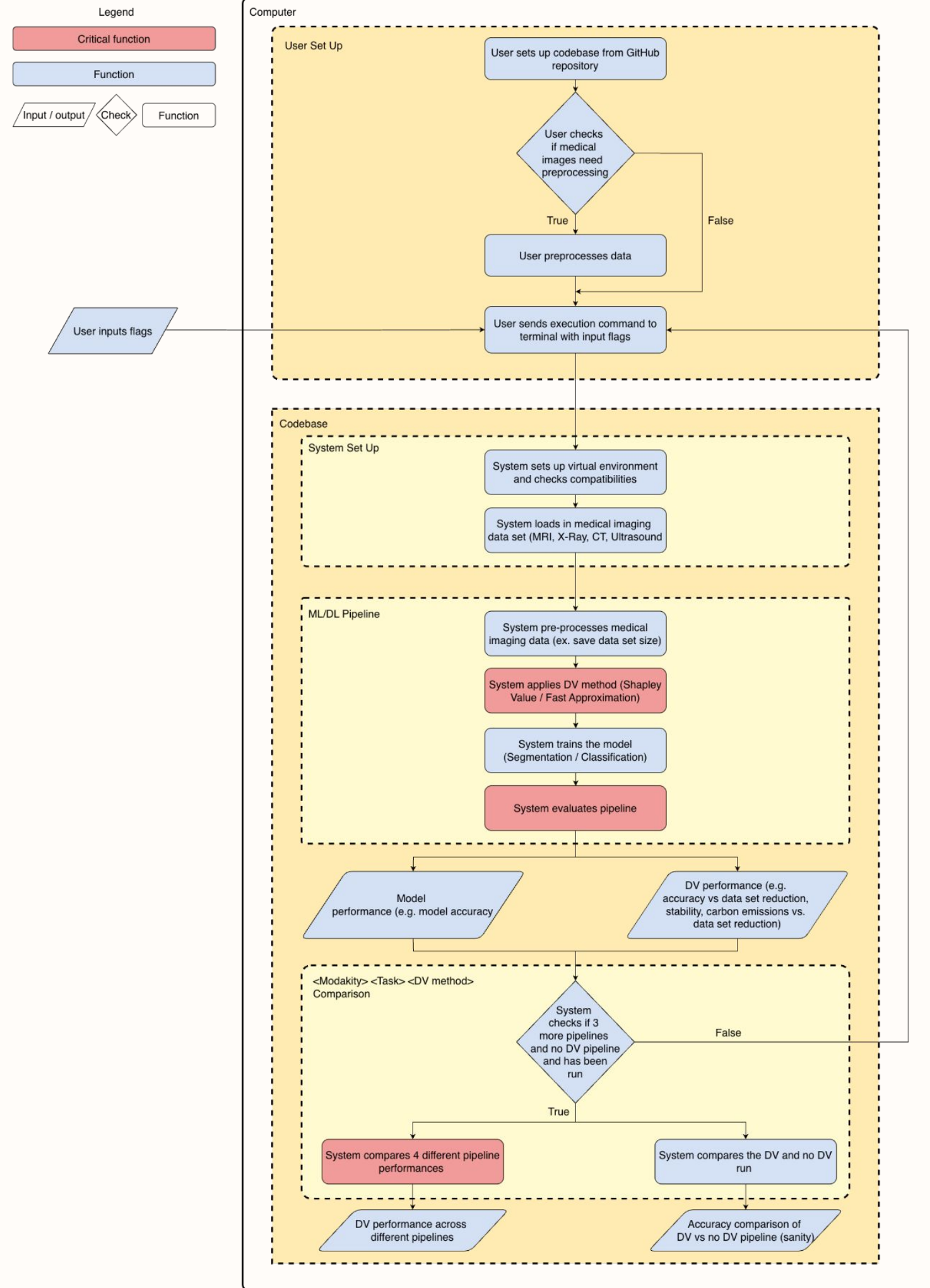
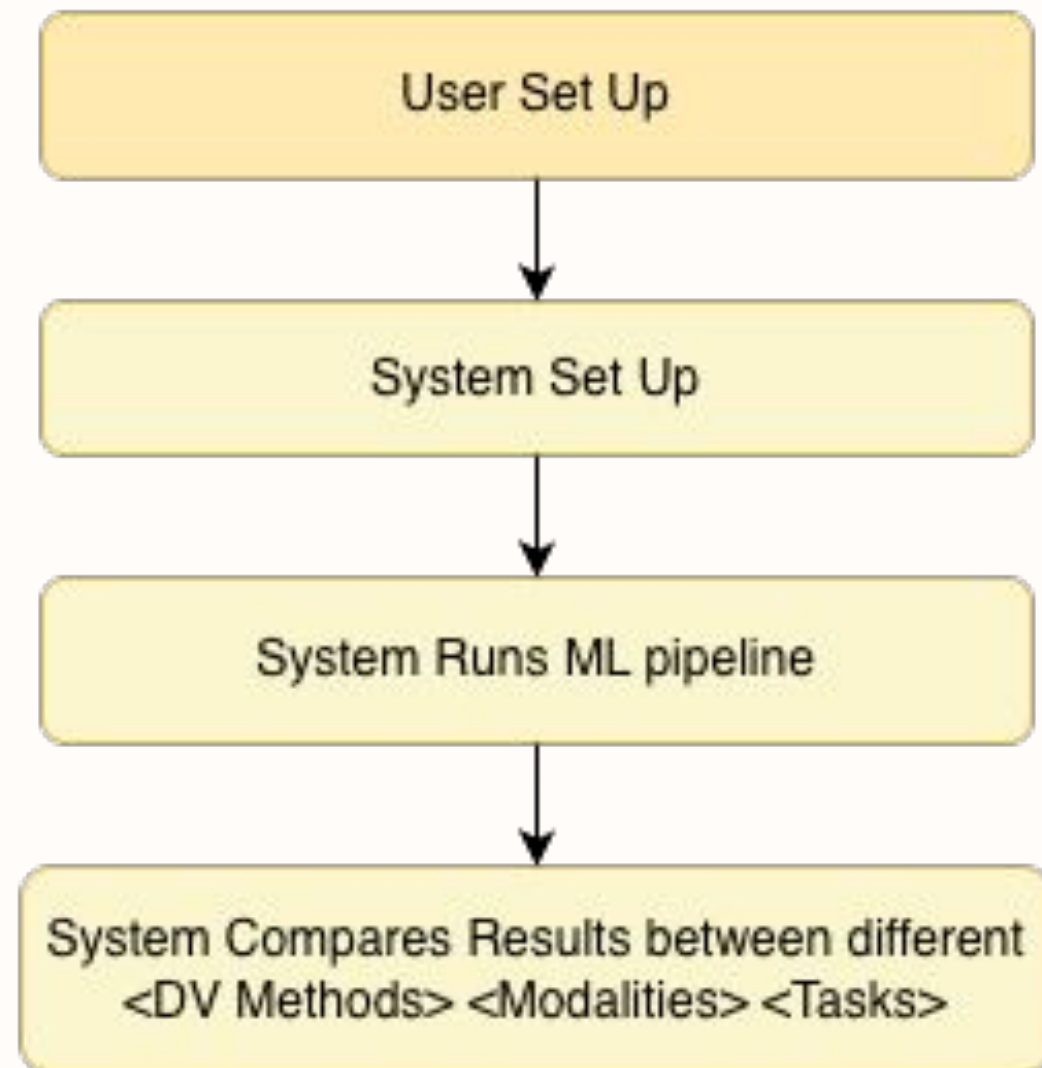
Can DV reduce current challenges in AI training and broaden application scope of ML use in medical imaging

Which tasks and modalities benefit most from data valuation?

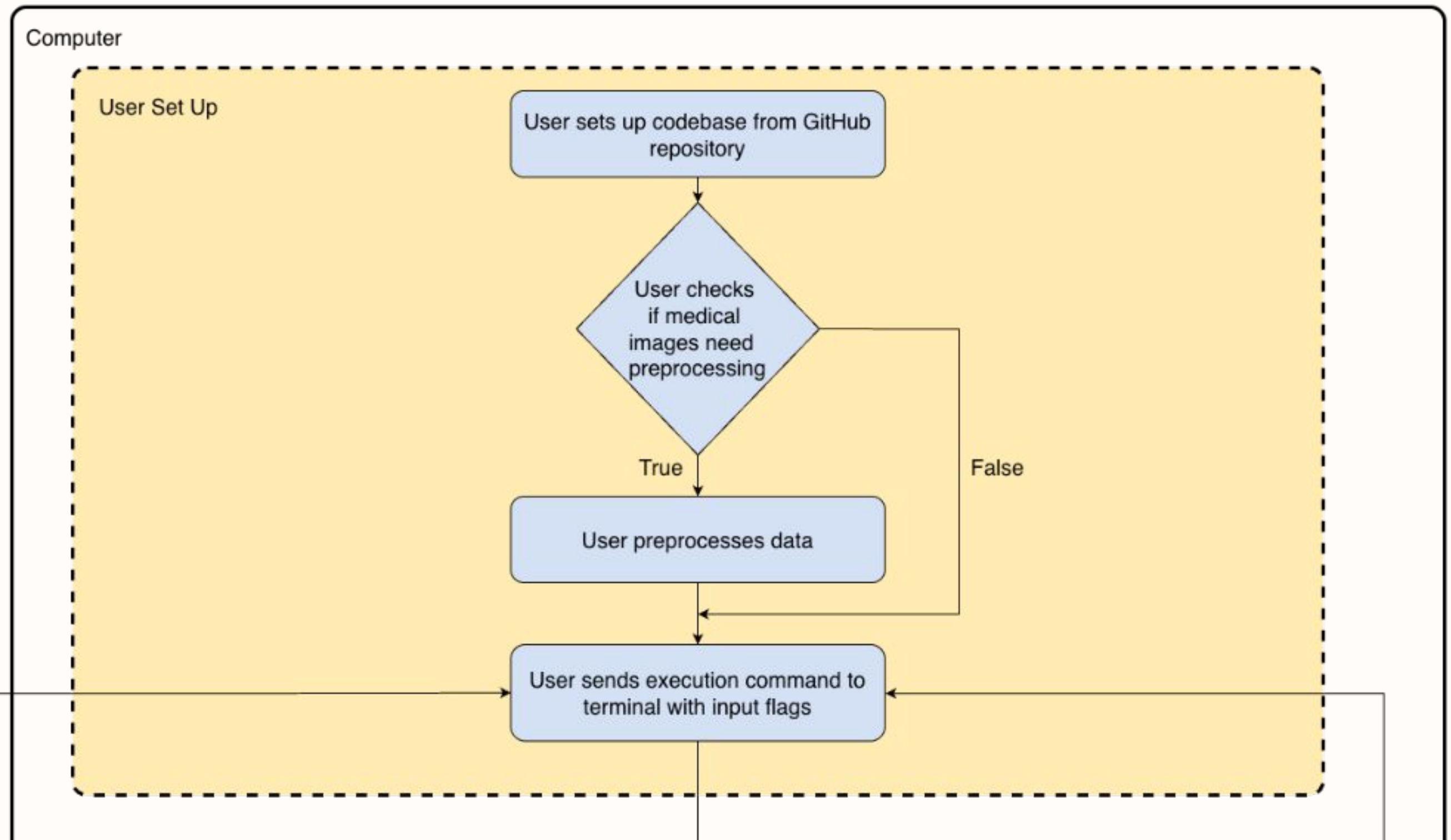
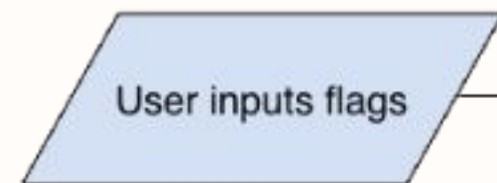
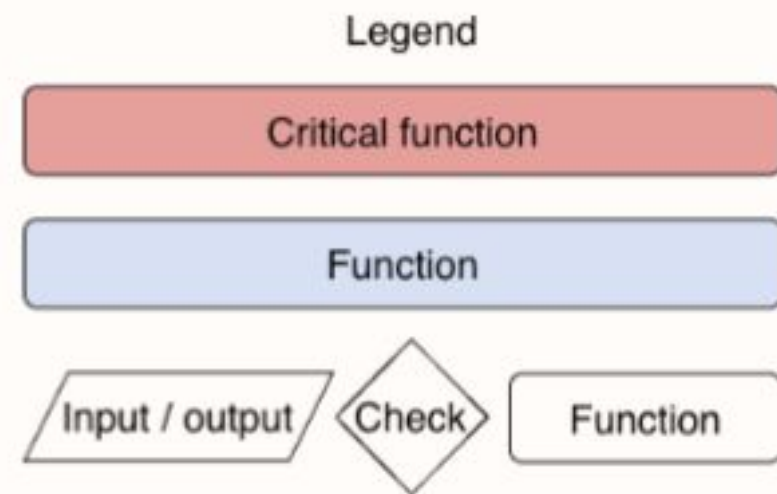
How can we balance data reduction and accuracy retention?

Function Structure Diagram (FSD)

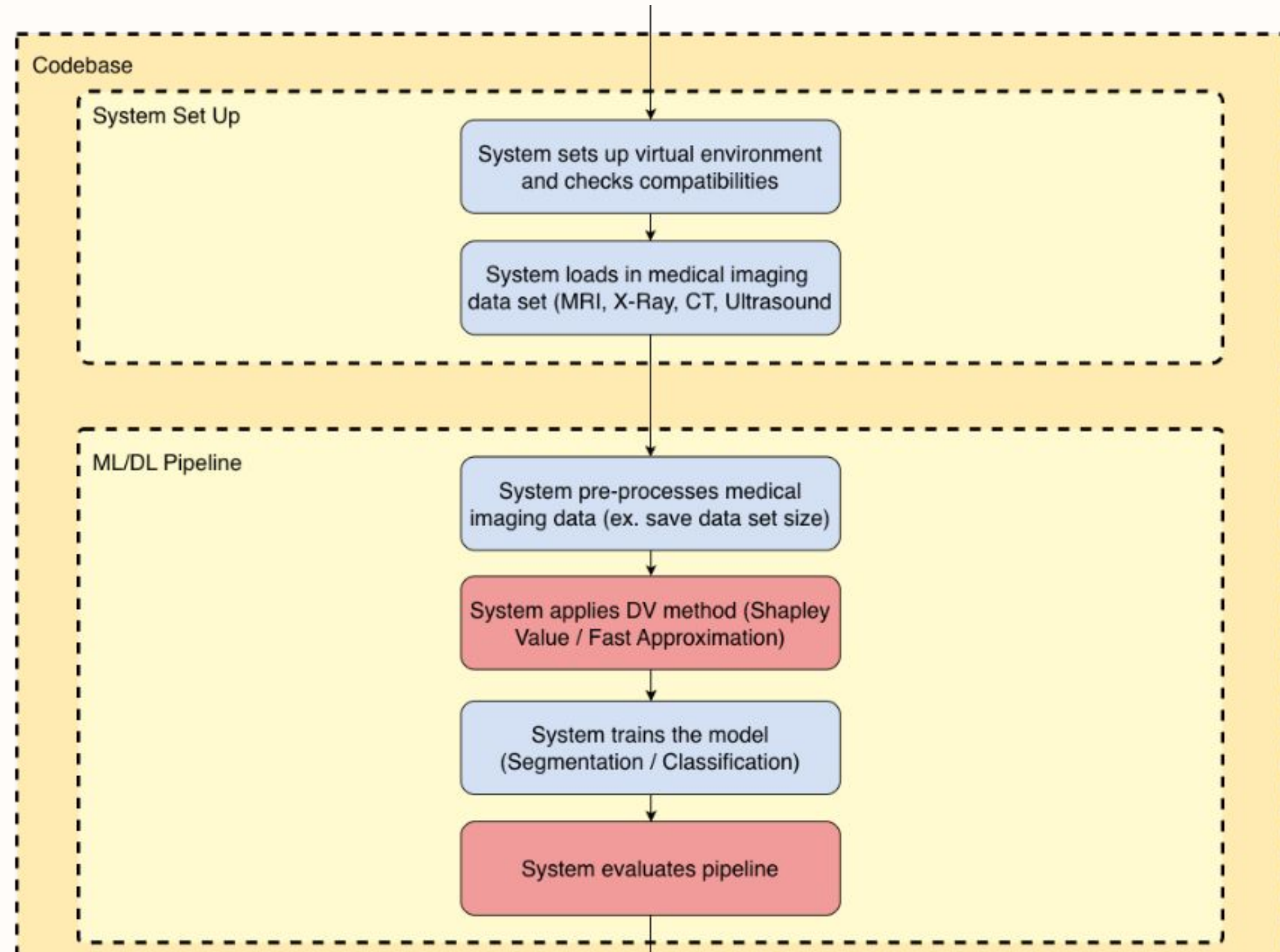
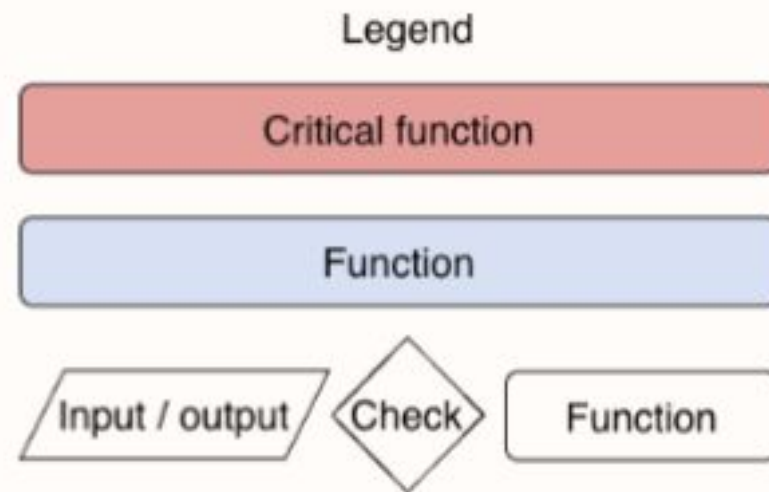
Function Structure Diagram Overview



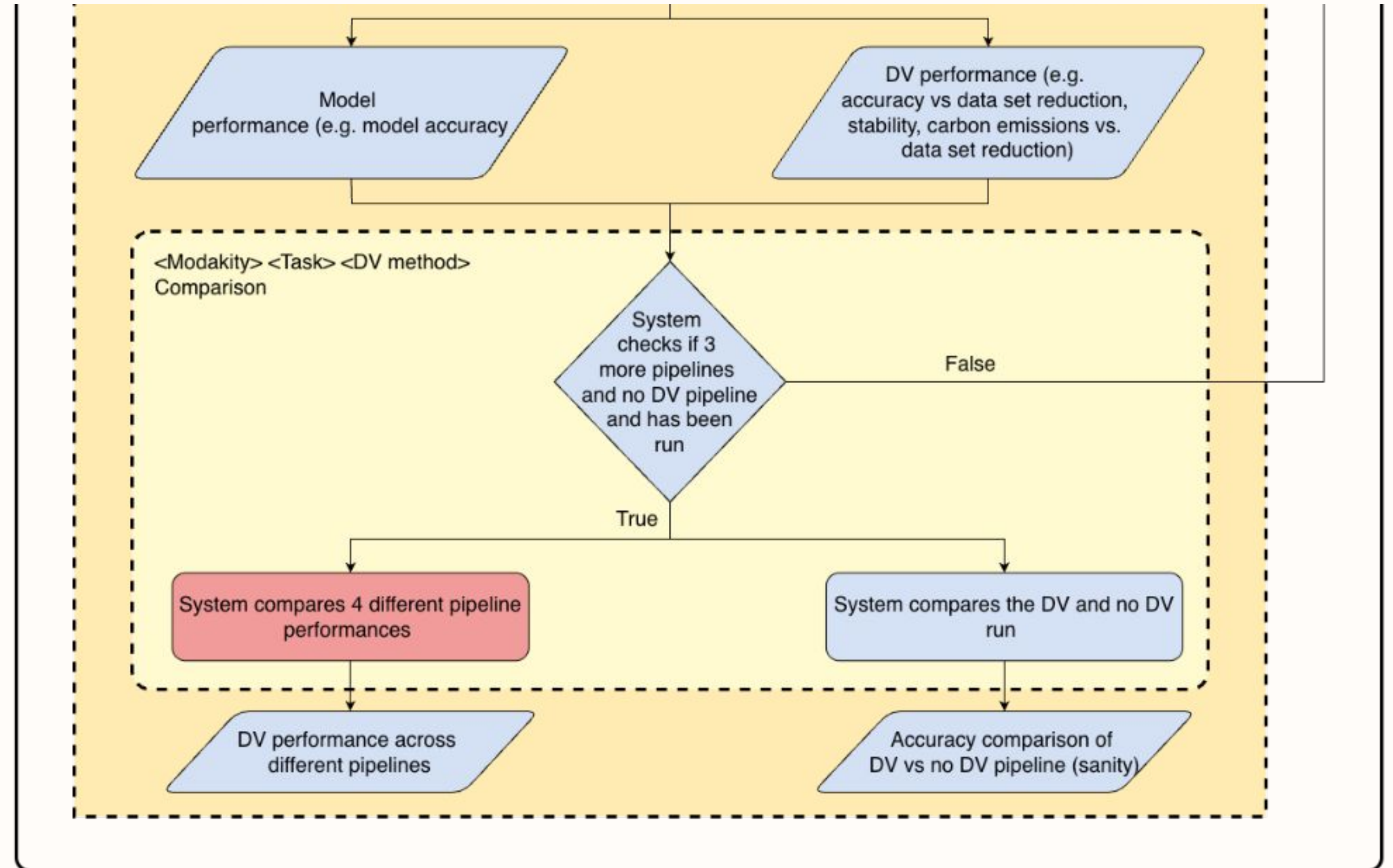
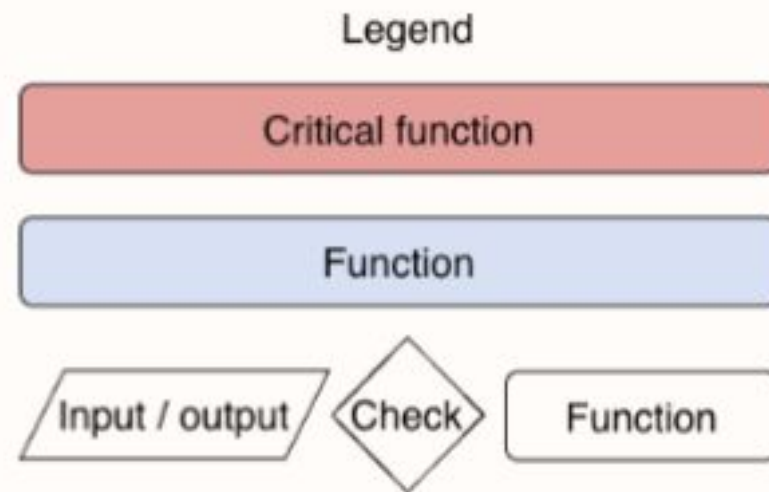
User Set Up



Codebase

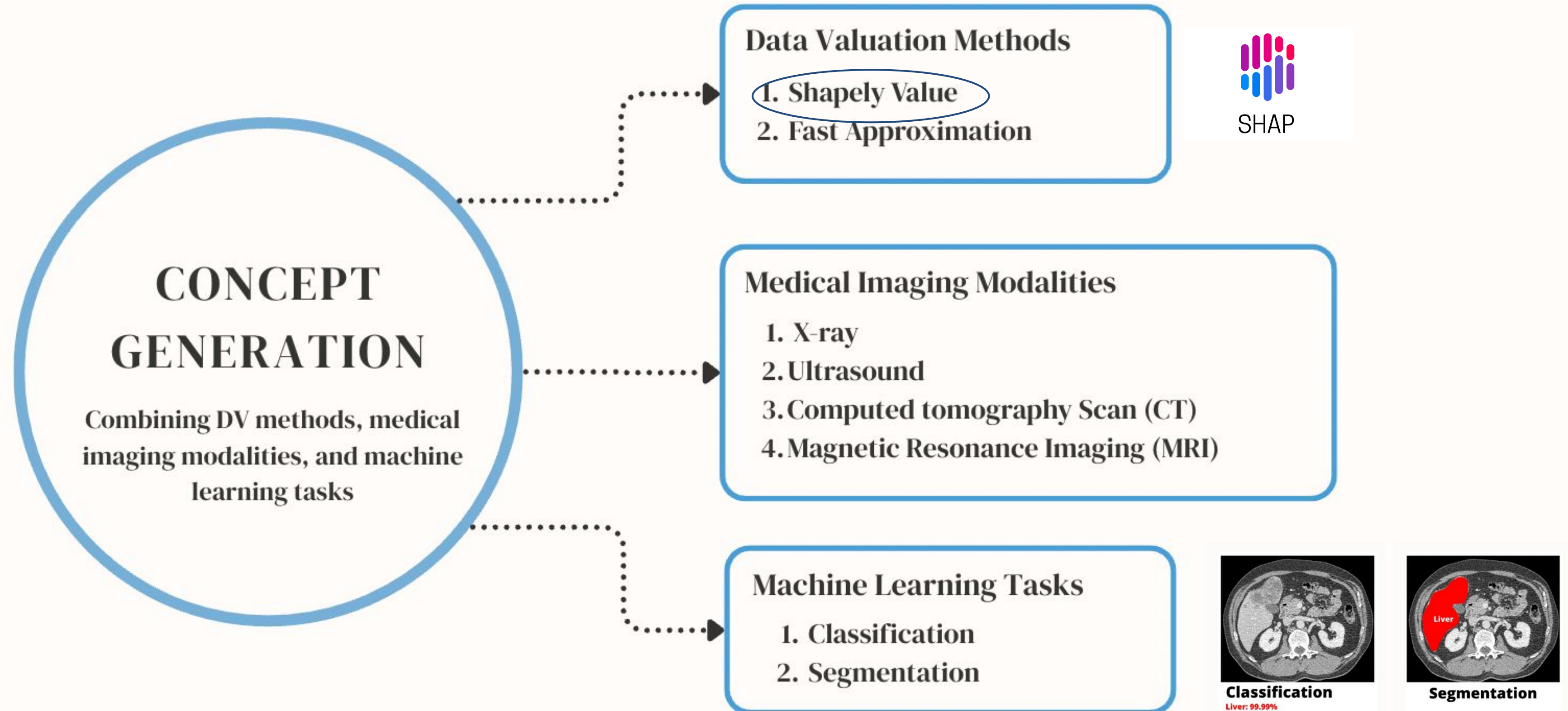


Codebase

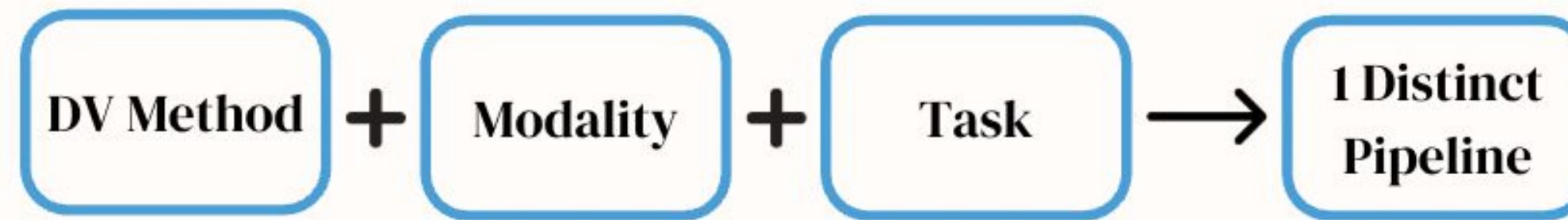


Concept Generation

Concept Generation



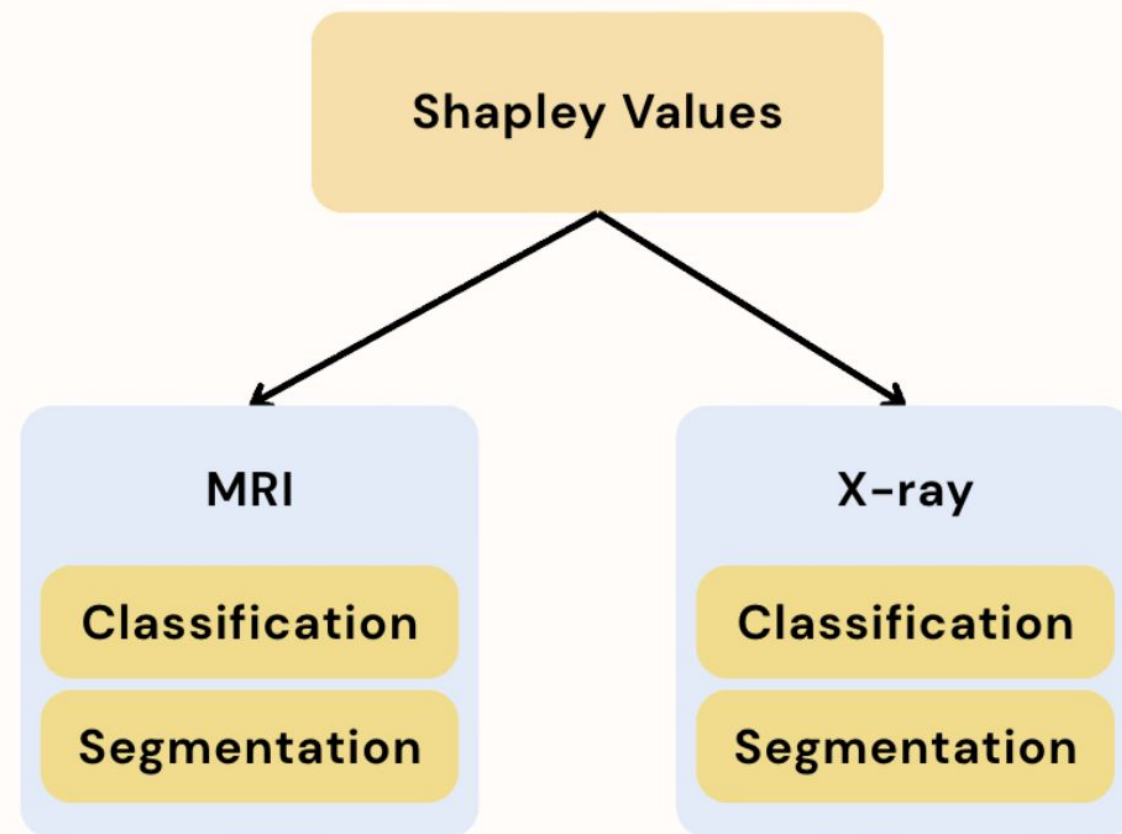
Concept Generation



Each Concept:

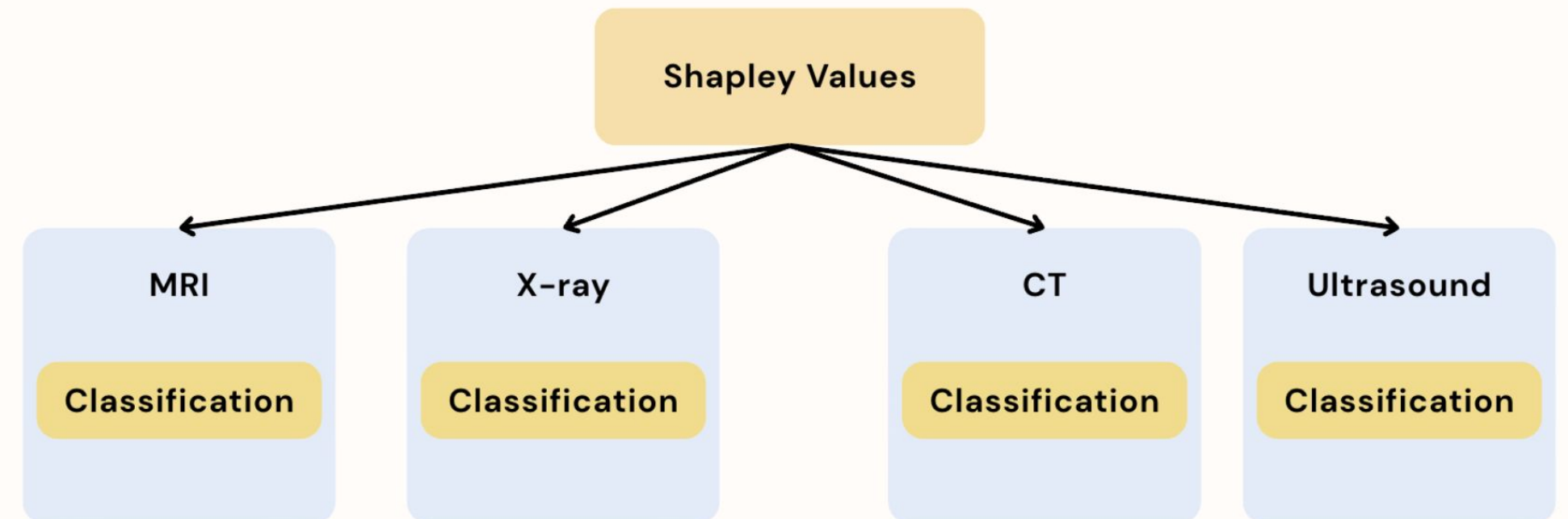
Combination of 4 Distinct Pipelines

Concept 1



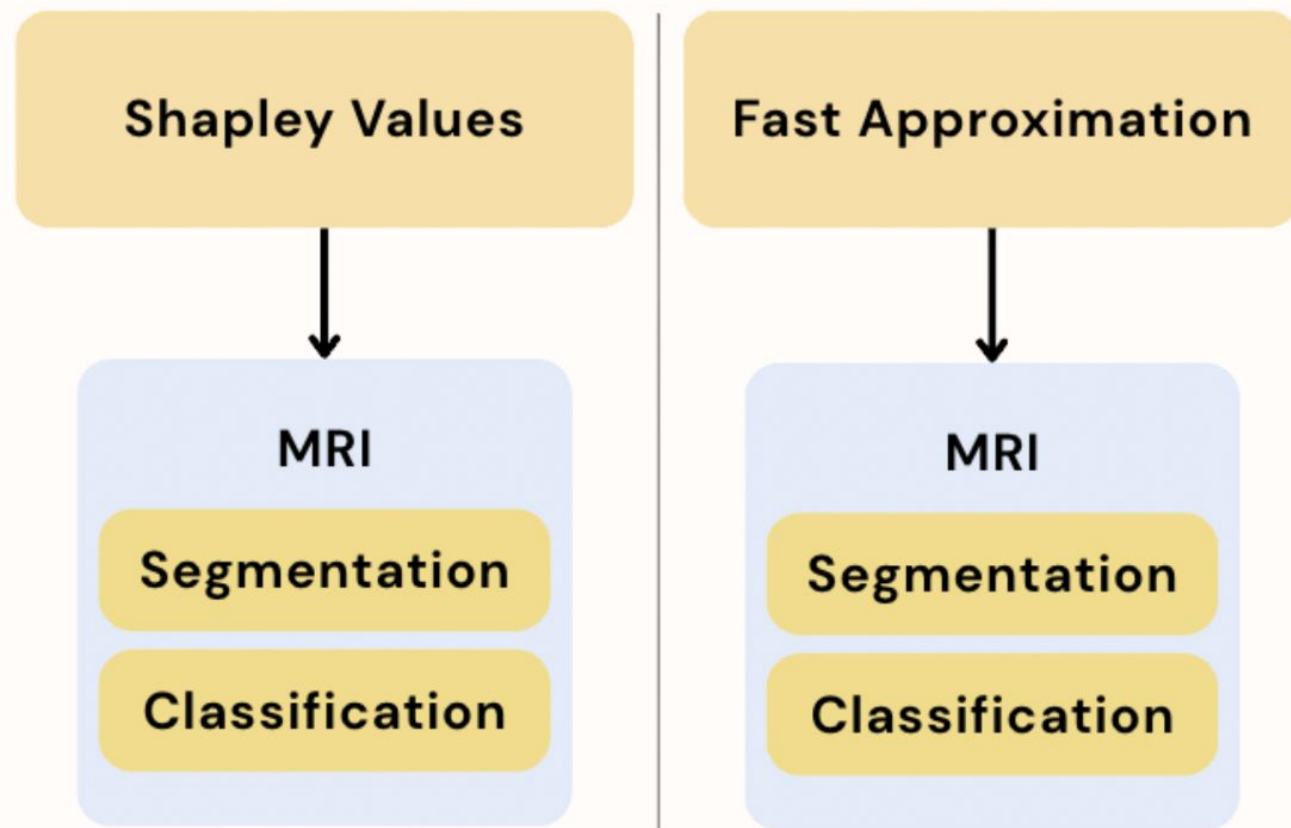
**Addresses Modality-Task Trade-offs
using Shapley Value**

Concept 2



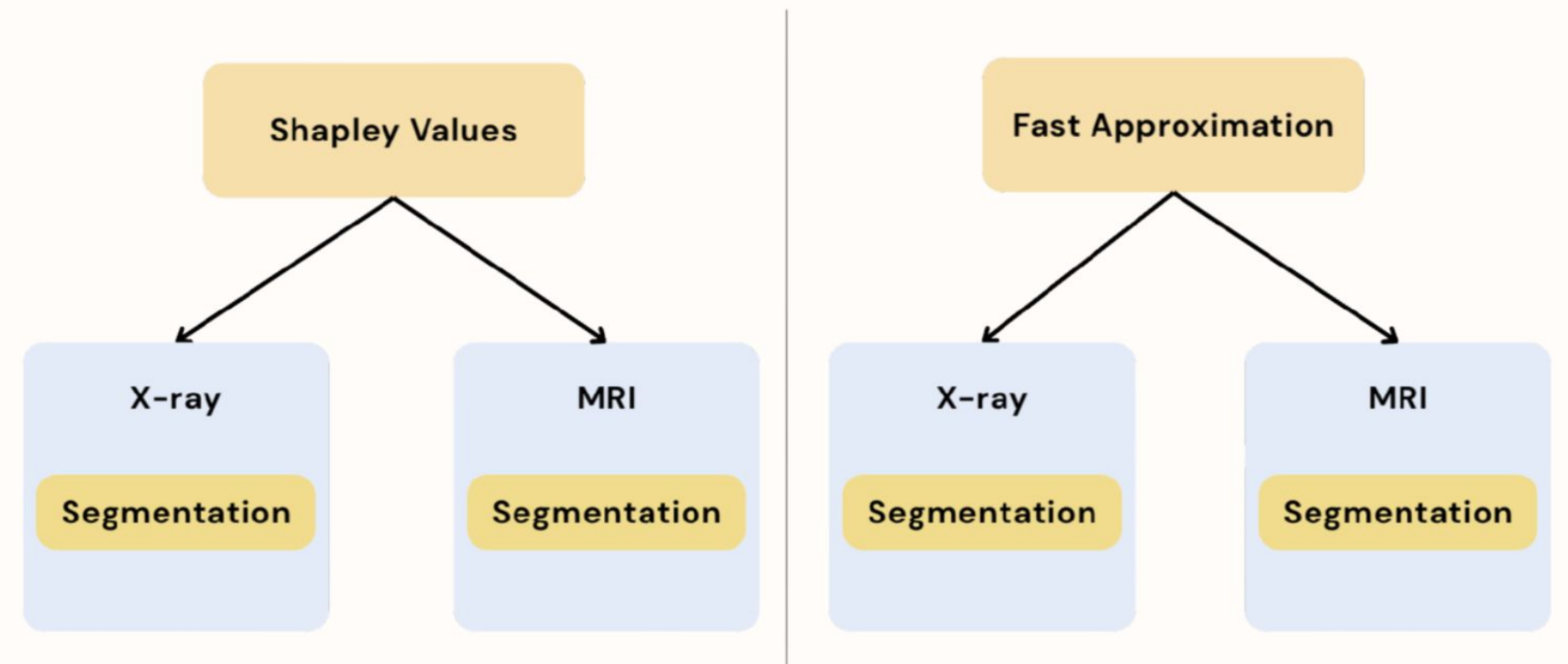
**Comparative Analysis of Four Modalities for
Classification using Shapley Value**

Concept 3



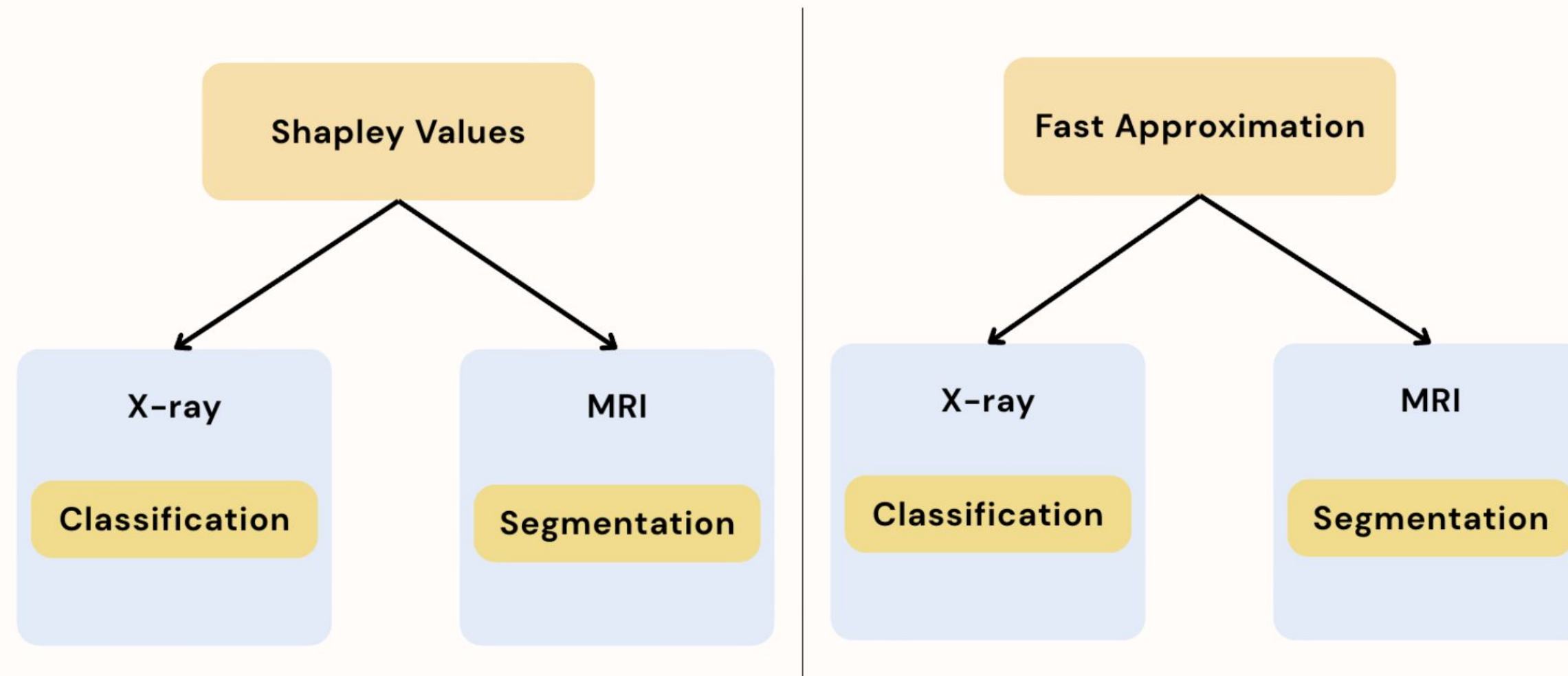
Thorough Analysis of DV Methods on One Modality

Concept 4



Highlights the Comparison of both DV Methods and Modalities on Segmentation

Concept 5



Comparative Data Valuation Across Modalities and Tasks

Concept Evaluation

Concept Screening and Scoring

Screening

- Only concepts with 4 pipelines passed to the scoring phase due to project scope and feasibility

Scoring Methodology

- 2 DV methods $\Rightarrow$ 100% score
- 4 imaging modalities $\Rightarrow$ 100% score
- 2 ML tasks $\Rightarrow$ 100% score

Evaluation Criteria Weighting

- Number of DV Methods (50%)
- Number of Imaging Modalities (25%)
- Number of ML Tasks (25%)

Concept Ranking

Concept	Rank	Score
1	4	0.675
2	4	0.675
3	3	0.6875
4	2	0.75
5	1	0.875

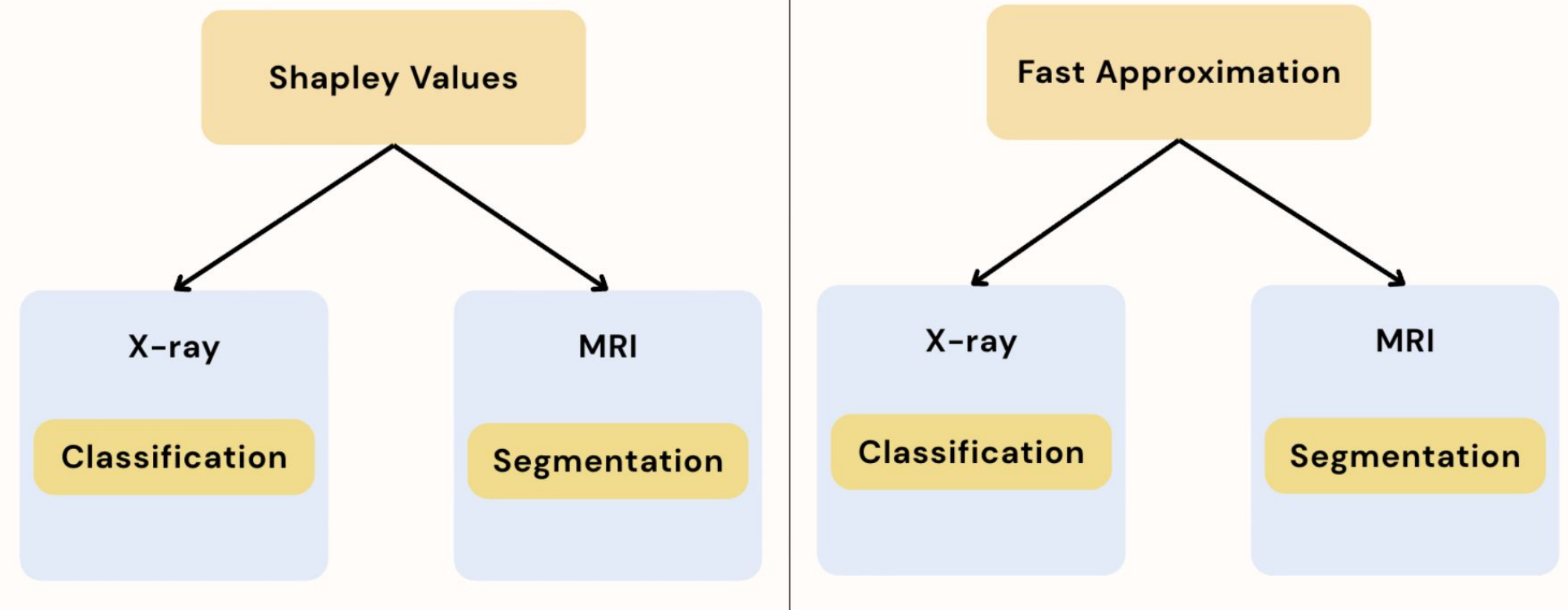
Conclusion and Final Concept

Strengths

- Maximum coverage allowing for thorough benchmarking of DV methods
- Comparable results between DV methods
- Compares exact and approximation methods for performance comparison

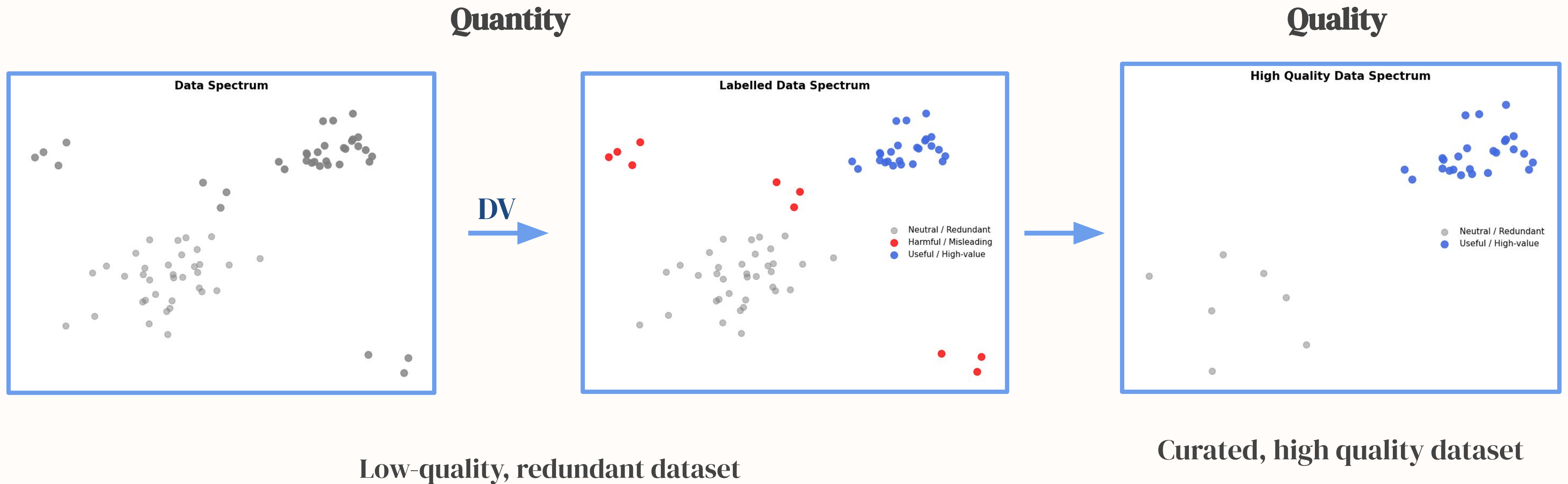
Next Steps

- Environment setup
- Pipeline implementation
- Performance benchmarking and analysis



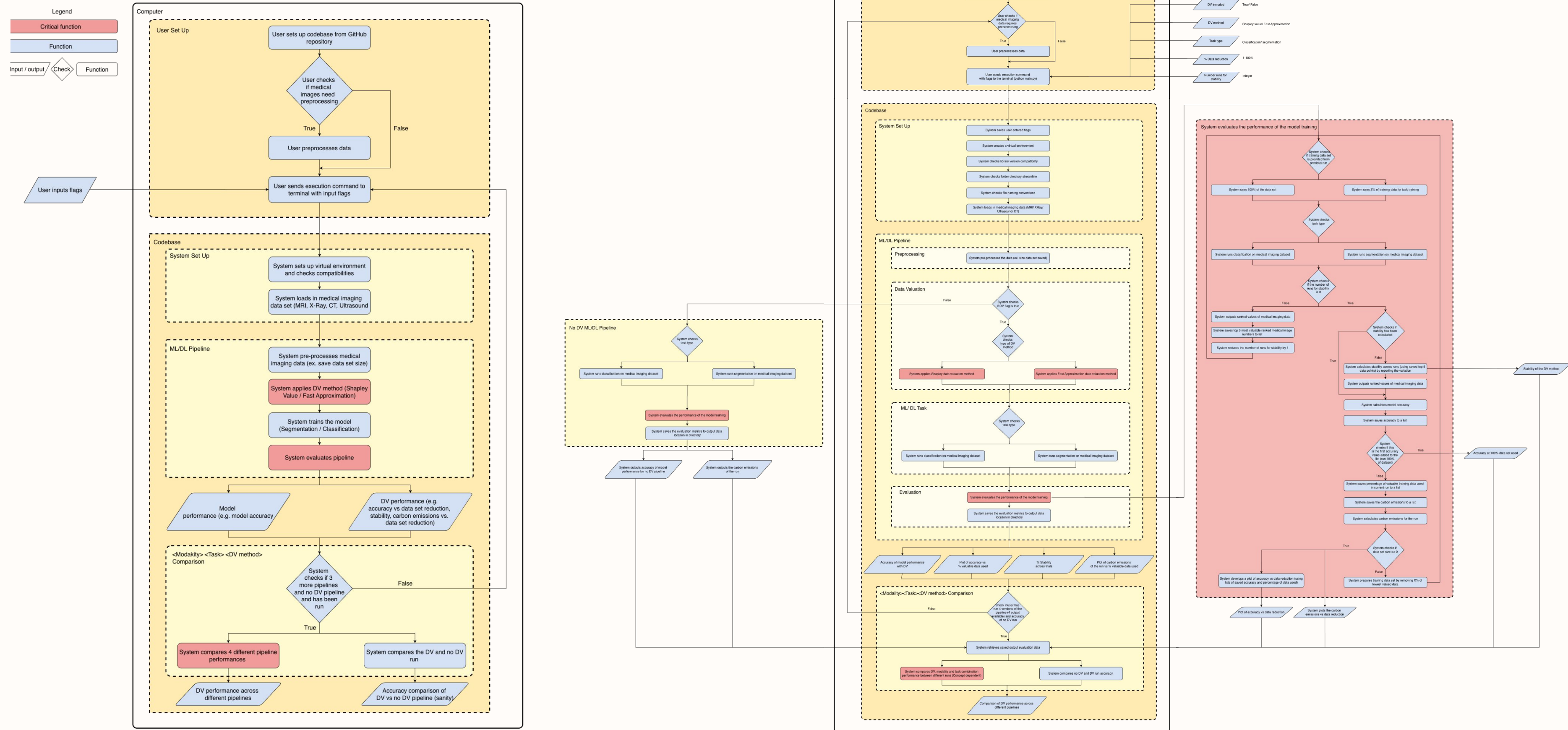
Appendix

From More Data ➡ To Better Data



Data valuation can help us identify useful, redundant, and harmful data points, then we can reduce size of dataset by removing harmful and redundant data points.

Function Structure Diagram



Weighted Decision Matrix

Concept	Number of DV Methods (50% Weight)	Number of Imaging Modalities (25% Weight)	Number of ML Tasks (25% Weight)	Rank	Score
1	1	2	2	4	0.675
2	1	4	1	4	0.675
3	2	1	2	3	0.6875
4	2	2	1	2	0.75
5	2	2	2	1	0.875

Accuracy Comparison of DV vs. no DV from Past Papers

Figure 2

